

Arthropods

PHYLUM	Arthropoda
CLASSES	19
ORDERS	123
FAMILIES	About 2,300
SPECIES	About 1.2 million

CLASSIFICATION NOTE

Arthropods are divided into 2 subphyla: chelicerates have pincerlike appendages and no antennae, while mandibulates have chewing mouthparts and one or 2 pairs of antennae.

MANDIBULATES
Hexapods (insects and other 6-legged arthropods) see pp.548–77
Centipedes and millipedes see p.578
Crustaceans see pp.579–84

CHELICERATES
Sea-spiders see p.585
Horseshoe crabs see p.585
Arachnids see pp.586–93

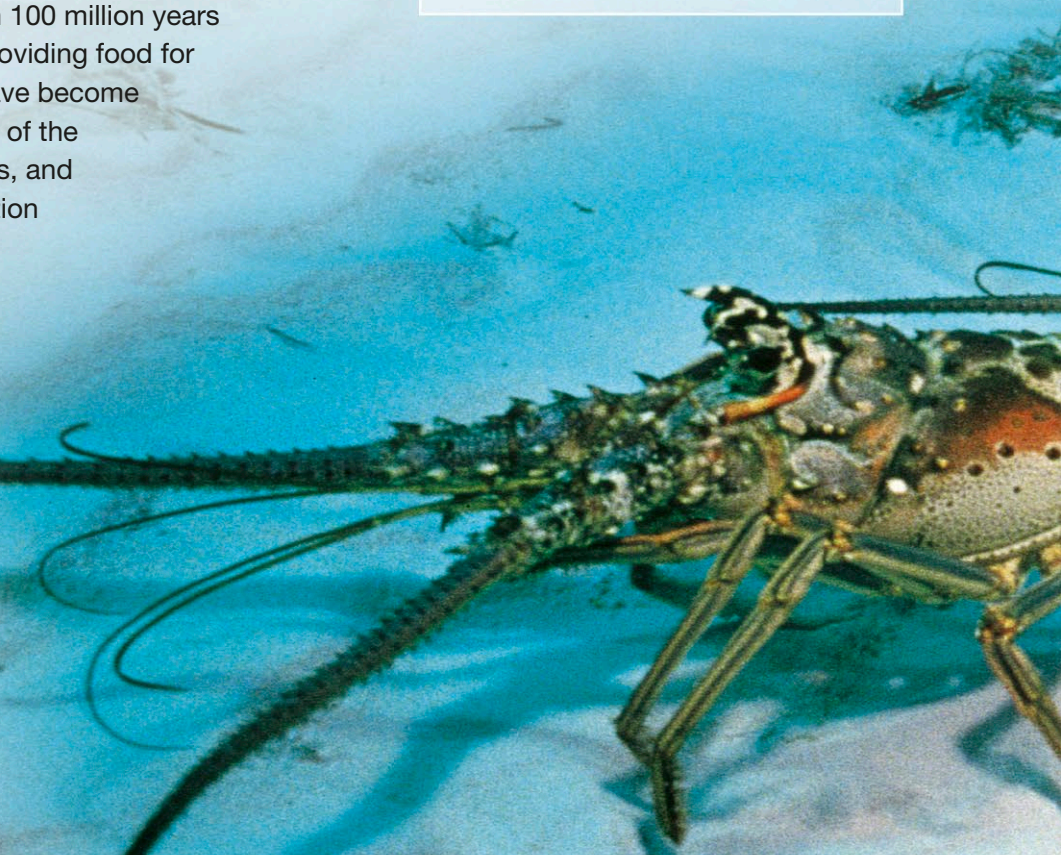
Arthropods form the largest phylum of living organisms and account for more than 3 out of 4 known species of animals. They include insects, centipedes and millipedes (myriapods), crustaceans, and arachnids, among others. All arthropods have an exoskeleton that covers a body divided into segments, and all have jointed legs. The first arthropods were marine animals, and many (especially crustaceans) are still found in the sea. However, insects, arachnids, and myriapods have successfully adapted to life on land. Insects, which are by far the most numerous of all arthropods, are the only invertebrates capable of powered flight, having evolved functional wings more than 100 million years before flying reptiles or birds. By providing food for many larger animals, arthropods have become essential to the functioning of most of the world’s ecosystems. Krill, copepods, and other crustaceans form the foundation of marine food chains, supporting

fish and marine mammals. On land, insects provide food for countless other animal species.

Anatomy

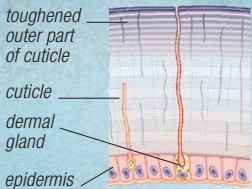
Arthropods share several common features. All species have a bilaterally symmetrical body that is divided into segments (see below). Arising from these segments are several jointed legs, which are arranged in pairs. The body is covered by a tough exoskeleton, which is produced by the epidermis and is composed of protein and a material called chitin. To allow for movement, the exoskeleton articulates at joints and hinges, where the cuticle is soft and flexible. In large marine species, the exoskeleton is strengthened by calcium carbonate, while land species have a thin layer of waterproof wax to stop them from drying out. All arthropods have an open circulatory system: their organs are bathed in a fluid called hemolymph, which is moved around the body by the heart. Gas exchange is carried out by gills, organs known as book lungs, or a system of tracheae. The nervous system consists of a brain that is connected via paired nerve cords to networks of nerve cells in the thorax and abdomen.

SEGMENTATION
The body of an arthropod is divided into segments. Arthropods probably evolved from marine worms that had unspecialized segments. Over time, eyes, antennae, and appendages appeared, and some segments were fused into functional units, of which the most common is the head. Primitive species, such as myriapods (a centipede is shown here), have a head and trunk. In arachnids, the head and thorax are fused to form a cephalothorax. Insects, the most advanced arthropods, have a head (made of 6 fused segments), a thorax (that has 3 segments), and an abdomen (with 11 segments).



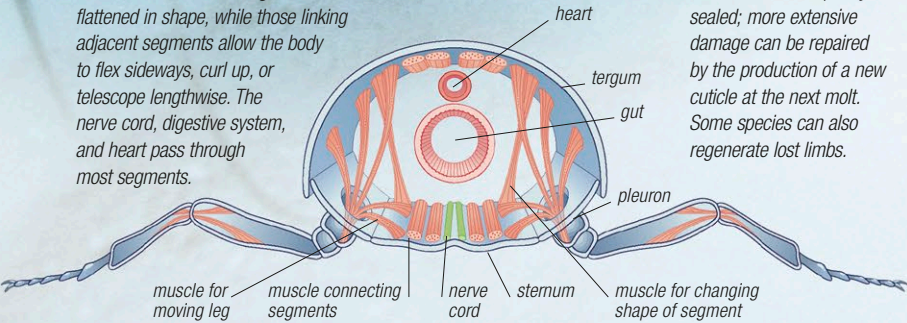
BODY SEGMENTS

Each segment (seen in cross section, below) is essentially a box, with a top (tergum), bottom (sternum), and sides (pleura). These outer surfaces are made from tough, relatively rigid plates (sclerites), which are joined by soft cuticle to allow for movement. Muscles between the tergum and sternum allow the segment to be flattened in shape, while those linking adjacent segments allow the body to flex sideways, curl up, or telescope lengthwise. The nerve cord, digestive system, and heart pass through most segments.



CUTICLE

The cuticle (seen in cross section, left) protects the body of an arthropod from damage and repels pathogens. Although it is a composite of several layers, the cuticle can be repaired: a small wound is quickly sealed; more extensive damage can be repaired by the production of a new cuticle at the next molt. Some species can also regenerate lost limbs.



MIGRATION

Arthropods are among the most mobile of invertebrates. Some species make use of this to migrate in response to changes in environmental conditions and food supply. Each year, many spiny lobsters make mass migrations between relatively warm and cool waters. Typically moving in single file, they maintain contact with one another by touch (using their antennae) or sight (picking out the light-colored spots on the abdomen of the individual in front of them). The fall migration is often triggered by violent storms.